

Department of Mining Engineering
Indian Institute of Technology (ISM), Dhanbad

Examination: MID SEMESTER

M. Tech (MNE & JRF) 1st Semester

Session: 2023-24

Subject: MNC 539 COMPUTATIONAL GEO MECHANICS AND GROUND CONTROL

Semester: Monsoon

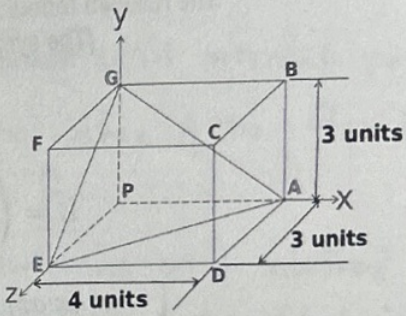
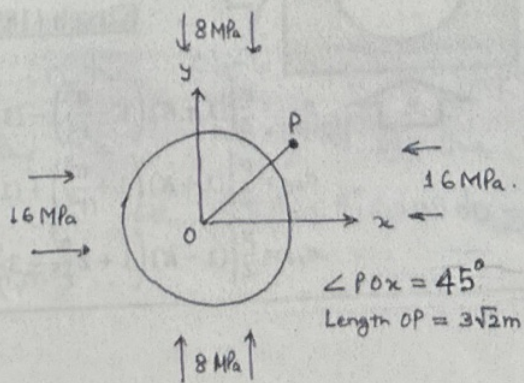
Instructions: 1. Answer ANY TWO questions

Time: 2 HOURS

2. Neat and accurate sketches will assist you in solving the problems logically and faster

Max. Marks: 30

3. Assume logical data, where necessary

Q. No.	Question	Marks
1	(a) Using neat sketches discuss the Cauchy's First and Second law of stress. (b) Using a neat sketch draw the components of traction acting on an element in two dimensions and show that $\tau_{xy} = \tau_{yx}$. (c) The state of stress at a point with respect to x-,y- and z-axis is given by the matrix $\begin{bmatrix} 15 & 10 & 5 \\ 10 & 30 & 15 \\ 5 & 15 & 5 \end{bmatrix} \text{ MPa}$ Determine the magnitude of normal and shear traction on the plane whose normal makes an angle of 45° with the x-axis and 60° with the y-axis.	6 3 6
2	(a) The rectangular state of stress at a point 'P' in a rock is given by the array $\begin{bmatrix} 5 & 5 & 3 \\ 5 & -5 & 4 \\ 3 & 4 & 8 \end{bmatrix}$ Determine the magnitude of the normal stress on the plane "AEG" shown in the figure on the right.  (b) The displacement field in rock mass is given by $\vec{u} = [7.5xz\hat{i} + 5yz\hat{j} + 2.5xy\hat{k}] \times 10^{-6} \text{ mm/mm}$ Determine the normal strain at a point (1, 1, 1) along a line AB, whose direction cosines $n_x = n_y = n_z = \frac{1}{\sqrt{3}}$ (c) A circular tunnel of 3.0 m radius is proposed to be excavated underground which is subjected to the <i>in situ</i> stress as shown in the figure below.  Determine the principal stresses at point P.	5 5 5

Q. No.	Question	Marks
3	(a) In case of infinitesimal strains in two dimensions. i. Define the normal and shear strains. ii. Using a neat and labeled sketch, prove that the normal strain in the x-direction is given by $\epsilon_{xx} = \frac{\partial}{\partial x}u(x, y)$ and the shear strain is given by $\epsilon_{xy} = \frac{1}{2} \left\{ \frac{\partial}{\partial y}u(x, y) + \frac{\partial}{\partial x}v(x, y) \right\}$. (b) The components of the state of stress at a point is given by the matrix $\begin{bmatrix} \sigma_x & 20 & 30 \\ 20 & 50 & 15 \\ 30 & 15 & -40 \end{bmatrix}$ Find the value of σ_x for which there exists a traction free plane and compute the direction cosines normal to the plane.	2+8 5

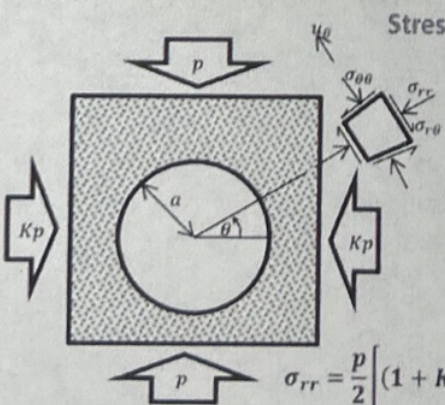
The relevant formula for solving the problem is given as under.
(The symbols have their usual meaning)

$$\tau' = L\tau L^T$$

$$L = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \text{ for 2D}$$

$$L = \begin{bmatrix} \cos(\angle x'ox) & \cos(\angle x'oy) & \cos(\angle x'oz) \\ \cos(\angle y'ox) & \cos(\angle y'oy) & \cos(\angle y'oz) \\ \cos(\angle z'ox) & \cos(\angle z'oy) & \cos(\angle z'oz) \end{bmatrix} \text{ for 3D}$$

Stress and displacement distribution around a circular opening



Kirsch (1898) Equations

$$\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4\frac{a^2}{r^2} + 3\frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + 3\frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + 2\frac{a^2}{r^2} - 3\frac{a^4}{r^4} \right) \sin 2\theta \right]$$



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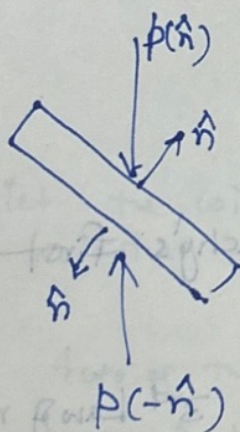
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Question No.	1	2	3	4	5	6	7	8	9	10	TOTAL
Marks Obtained											

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Ques-1(a): Cauchy's First Law of Stress

Consider a thin penny-shaped slab of rock having thickness 'h' and radius 'r'.



If the outward unit normal vector on the right face of the slab is denoted by $\hat{n}$

Then the outward unit normal vector on the left face of the slab is $-\hat{n}$.

$\therefore$ The total force acting on the face with outward unit normal vector $\hat{n}$ is $= \pi r^2 p(\hat{n})$.

Similarly on the opposite face it is $= \pi r^2 p(-\hat{n})$

The total force acting on the outer rim of the penny shaped slab will be given by the integral of the traction over the outer area and will be proportional to $2\pi r h$ (Surface area of the rim).

Performing the force balance on the slab yields

$$\pi r^2 p(\hat{n}) + \pi r^2 p(-\hat{n}) + 2\pi r h \vec{t} = 0$$

Where $\vec{t}$ is the mean traction over the outer Rim.

if the thickness h of the slab is allowed to vanish, the third term will vanish, so the condition for equilibrium will be

$$\pi r^2 p(\hat{n}) + \pi r^2 p(-\hat{n}) = 0$$

$$\Rightarrow \boxed{\vec{p}(-\hat{n}) = -p(\hat{n})} \quad \text{--- (1)}$$

The equation (1) is known as "Cauchy's first law of stress".

This law is an analogue of Newton's 3rd Law of motion.

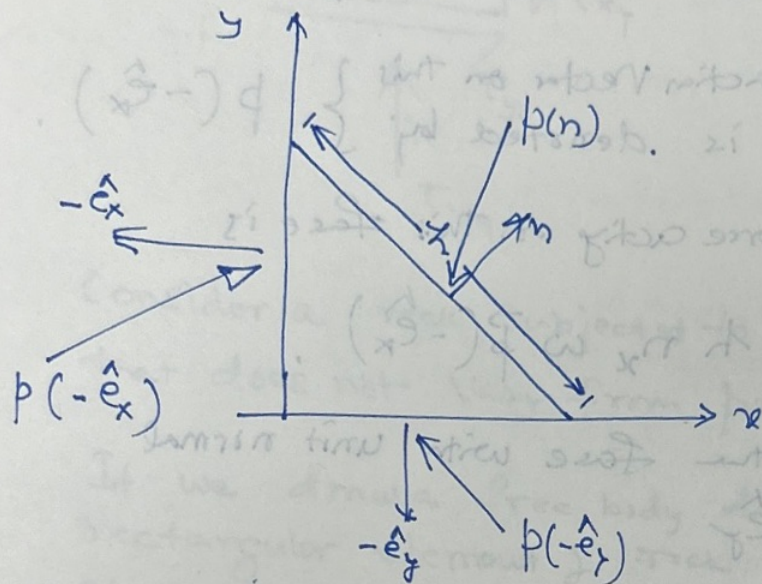
"If the material to the left of a given plane exerts a traction $\vec{p}$ on the material on the right then the material on the right will exert a traction $-\vec{p}$ on the material to the left."

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3

Cauchy's 2nd Law of Stress

Consider a triangular slab of rock as shown in the figure below.



Let the width of the slab in the z-direction $= w$.

Two of the faces of this slab have outward unit normal vector that co-incides with the negative x & y coordinate directions respectively.

The third face has an outward unit normal $\hat{n} = \langle n_x, n_y \rangle$.

Let the length of the face with outward normal $\hat{n}$ $= h$.

The length of face that has
outward unit normal $-\hat{e}_x$ is $h \hat{n}_x$

$$\text{and its area} = h \hat{n}_x w.$$

also the traction vector on this
face is denoted by $\vec{p}(-\hat{e}_x)$.

$\therefore$ The total force acting on this face is

$$= h n_x w \vec{p}(-\hat{e}_x).$$

Similarly for the face with unit normal
vector $-\hat{e}_y$

The total force acting on the
said face is $h w \hat{n}_y \vec{p}(\hat{e}_y)$.

Now, the Force Balance on this slab leads to

$$h w n_x \vec{p}(-\hat{e}_x) + h w \hat{n}_y \vec{p}(-\hat{e}_y)$$

$$+ h w \vec{p}(\hat{n}) = 0$$

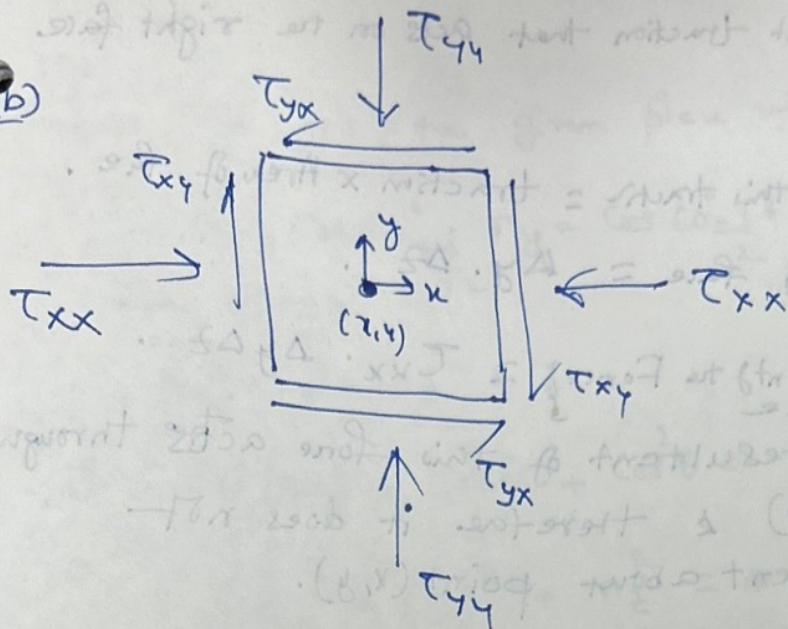
Cancelling out the common terms and

using the Cauchy's 1st Law we

have

$$\boxed{\vec{p}(\hat{n}) = n_x \vec{p}(\hat{e}_x) + n_y \vec{p}(\hat{e}_y)}$$

5



Consider a rock subjected to a state of stress, that does not vary from point to point.

If we draw a free body diagram for a small rectangular element of rock ~~subje~~ centred at a point (x, y) , the traction components acting on the four faces can be shown as in the figure above.

Let Length of element in x -direction = Δx
 — — — — — y -direction = Δy
 — — — — — z -direction = Δz

In order for the element to be in equilibrium, the sum of moments about any point say (x, y) must be zero.

$$\tau_{xy} = \tau_{yx}$$

- ② Consider the first traction that acts on the right face of the element.

The force vector of this traction = traction $\times$ Area of face.

$$\text{Here Area of face} = \Delta y \cdot \Delta z.$$

$$\therefore \text{The x-component of the Force on this face} = \tau_{xx} \cdot \Delta y \Delta z.$$

However, the resultant of this force acts through the centre (x, y) & therefore it does not exert any moment about point (x, y) .

$$\text{Now, the y-component of this traction on the said face} = \tau_{xy}.$$

$$\text{The net force associated with it} = \tau_{xy} \Delta y \Delta z.$$

$$\text{The moment arm of the force} = \frac{\Delta x}{2}.$$

$\therefore$ For the Right face

The total moment about the z-axis through the point (x, y)

$$= \tau_{xy} \cdot \frac{\Delta x \cdot \Delta y \Delta z}{2}.$$

Therefore, adding up all the four moments that are contributed by the four Shear Stresses fields:

$$\tau_{xy} \cdot \frac{\Delta x \Delta y \Delta z}{2} + \tau_{xy} \cdot \frac{\Delta x \Delta y \Delta z}{2} - \tau_{yx} \cdot \frac{\Delta x \Delta y \Delta z}{2} - \tau_{yx} \cdot \frac{\Delta x \Delta y \Delta z}{2} = 0$$

on simplification it yields.

$$\boxed{\tau_{xy} = \tau_{yx}}$$

Ans: (c)

(7)

The dir's of the given plane will be

$$n_x = \cos 45^\circ = \frac{1}{\sqrt{2}}, \quad n_y = \cos 60^\circ = \frac{1}{2}, \quad n_z = \cos \theta.$$

But since, $\hat{n}^2 = 1$

$$n_x^2 + n_y^2 + n_z^2 = 1$$

$$\Rightarrow \cos^2(45) + \cos^2 60 + \cos^2 \theta = 1$$

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + n_z^2 = 1$$

$$n_z^2 = 1 - \frac{1}{2} - \frac{1}{4} = \frac{4 - 2 - 1}{4} = \frac{1}{4}$$

$$\Rightarrow n_z = \frac{1}{2}$$

$$\hat{n} = \left(\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{2} \right)$$

$$\sigma = \tau \hat{n} \cdot \hat{n}$$

$$= \begin{Bmatrix} 15 & 10 & 5 \\ 10 & 30 & 15 \\ 5 & 15 & 5 \end{Bmatrix}_{3 \times 3} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}_{3 \times 1} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}_{1 \times 3}$$

$$= \begin{bmatrix} 18.1066 \\ 29.5711 \\ 13.5355 \end{bmatrix}^T \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = 34.357 \text{ MPa}$$

$$\tau = \sqrt{|\rho||\rho| - \sigma^2} = \sqrt{(18.1066)^2 + (29.5711)^2 + (13.5355)^2 - (34.357)^2}$$

$$= 14.32 \text{ MPa.}$$

⑦
2(a)

$$\vec{EA} = \vec{PA} - \vec{PE}$$

$$= 4\hat{i} - 3\hat{k}$$

$$\vec{EG} = \vec{PG} - \vec{PE} = 3\hat{j} - 3\hat{k}$$

$$\vec{EA} \times \vec{EG} = (4\hat{i} - 3\hat{k}) \times (3\hat{j} - 3\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 0 & -3 \\ 0 & 3 & -3 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} 0 & -3 \\ 3 & -3 \end{vmatrix} + (-\hat{j}) \begin{vmatrix} 4 & -3 \\ 0 & -3 \end{vmatrix} + \hat{k} \begin{vmatrix} 4 & 0 \\ 0 & 3 \end{vmatrix}$$

$$= 9\hat{i} + 12\hat{j} + 12\hat{k}$$

The vector $\vec{EA} \times \vec{EG}$ is $\perp^r$ to the plane AEG

or, The vector $3(3\hat{i} + 4\hat{j} + 4\hat{k})$ is also $\perp^r$ to AEG.

$\therefore$ The unit vector $\perp^r$ to the plane AEG after normalization is

$$\hat{n} = \frac{3\hat{i} + 4\hat{j} + 4\hat{k}}{\sqrt{3^2 + 4^2 + 4^2}}$$

$$= \frac{3}{\sqrt{41}}\hat{i} + \frac{4}{\sqrt{41}}\hat{j} + \frac{4}{\sqrt{41}}\hat{k}$$

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The traction on the Given plane is therefore

$$\vec{P}(\hat{n}) = \tau \hat{n}$$

$$= \frac{1}{\sqrt{41}} \begin{bmatrix} 5 & 5 & 3 \\ 5 & -5 & 4 \\ 3 & 4 & 8 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix}$$

$$= \frac{1}{\sqrt{41}} \begin{bmatrix} 47 \\ 11 \\ 57 \end{bmatrix}$$

The Normal stress = $\vec{P}(\hat{n}) \cdot \hat{n}$

$$= \frac{1}{\sqrt{41}} \begin{bmatrix} 47 & 11 & 57 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} \times \frac{1}{\sqrt{41}}$$

$$= \frac{1}{41} \begin{bmatrix} 47 & 11 & 57 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} = 10.073 \text{ MPa.}$$

$$\left\{ \begin{matrix} \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{matrix} \right\} = \frac{1}{2} \left\{ \begin{matrix} \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{matrix} \right\}$$

2(b) $\vec{u} = [7.5xz \hat{i} + 5yz \hat{j} + 2.5xy \hat{k}] \times 10^{-6}$

$$u = 7.5xz \times 10^{-6}$$

$$v = 5yz \times 10^{-6}$$

$$w = 2.5xy \times 10^{-6}$$

$$\vec{\nabla u} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{bmatrix} 7.5z & 0 & 7.5x \\ 0 & 5z & 5y \\ 2.5y & 2.5x & 0 \end{bmatrix} \times 10^{-6}$$

$$(\vec{\nabla u})_{(1,1,1)} = \begin{bmatrix} 7.5 & 0 & 7.5 \\ 0 & 5 & 5 \\ 2.5 & 2.5 & 0 \end{bmatrix} \times 10^{-6}$$

$$\epsilon_{(1,1,1)} = \frac{1}{2} \left\{ \vec{\nabla u} + (\vec{\nabla u})^T \right\}$$

(11)

$$= \frac{1}{2} \left[\begin{vmatrix} 7.5 & 0 & 7.5 \\ 0 & 5 & 5 \\ 2.5 & 2.5 & 0 \end{vmatrix} + \begin{vmatrix} 7.5 & 0 & 2.5 \\ 0 & 5 & 2.5 \\ 7.5 & 5 & 0 \end{vmatrix} \right]$$

$$= \frac{1}{2} \begin{vmatrix} 15 & 0 & 10 \\ 0 & 10 & 7.5 \\ 10 & 7.5 & 0 \end{vmatrix} \times 10^{-6}$$

$$= \begin{vmatrix} 7.5 & 0 & 5 \\ 0 & 5 & 3.75 \\ 5 & 3.75 & 0 \end{vmatrix} \times 10^{-6}$$

$$\epsilon_n = \left\{ \begin{bmatrix} \epsilon \end{bmatrix}_{3 \times 3} \begin{bmatrix} \hat{n} \end{bmatrix}_{3 \times 1} \right\}^T \begin{bmatrix} \hat{n} \end{bmatrix}_{3 \times 1}$$

$$= 10 \times 10^{-6}$$

$$\sigma = 2\epsilon - (\epsilon - \sigma)(\sigma - \epsilon) \in P \sigma = \begin{vmatrix} 2 - (\sigma - \epsilon) \\ (\sigma - \epsilon) & 2 \end{vmatrix}$$

$$\sigma = \epsilon \sigma + \sigma \epsilon - \epsilon^2 \Rightarrow \sigma = 2\epsilon - \sigma \epsilon + \sigma \epsilon - \epsilon^2 \Rightarrow$$

$$\begin{vmatrix} \sigma - 1.81 \text{ MPa} \\ \sigma - 1.18 \text{ MPa} \end{vmatrix} = 0$$

11
Ans- 2(c)

$$p = 8 \text{ MPa}$$

$$kp = 16 \text{ MPa} \Rightarrow k = 2$$

$$a = 3$$

$$r = 3\sqrt{2}$$

$$r^2 = 3^2 \times 2$$

$$r^4 = 3^4 \times 2^2$$

$$\sigma_{rr} = \frac{p}{2} \left[(1+2) \left(1 - \frac{3 \times 3}{3^2 \times 2} \right) \right] = 4 \times 3 \times \left(1 - \frac{1}{2} \right) = 6 \text{ MPa}$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+2) \left(1 + \frac{3 \times 3}{3^2 \times 2} \right) \right] = 4 \times 3 \times \frac{3}{2} = 18 \text{ MPa}$$

$$\begin{aligned} \sigma_{r\theta} &= \frac{p}{2} \left[(1-2) \left(1 + \frac{2 \times 3^2}{3^2 \times 2} - \frac{3 \times 3^4}{3^4 \times 2^2} \right) \right] \\ &= 4 \times (-1) \left(1 + 1 - \frac{3}{4} \right) = -\frac{4 \times 5}{4} = -5 \text{ MPa} \end{aligned}$$

$\therefore$ The stress matrix at point P is

$$\begin{bmatrix} 6 & -5 \\ -5 & 18 \end{bmatrix}$$

The principal stress can be found by eigen value of the matrix.

$$\begin{vmatrix} (6-\sigma) & -5 \\ -5 & (18-\sigma) \end{vmatrix} = 0 \Rightarrow (\sigma-18)(\sigma-6) - 25 = 0$$
$$\Rightarrow \sigma^2 - 24\sigma + 108 - 25 = 0 \Rightarrow \sigma^2 - 24\sigma + 83 = 0$$

$$\Rightarrow \sigma_1 = 19.81 \text{ MPa}$$

$$\sigma_2 = 4.18 \text{ MPa}$$

$\Rightarrow$ These are the principal stresses.

DHANBAD - 826004, (JHARKHAND), INDIA
मध्यावदी परीक्षा (Mid Semester Examination) 13

Adm. No. : _____

Part : _____ Semester : _____

Name of Invigilator : _____ Date : _____

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Marks Obtained :											

Signature of Examiner : _____

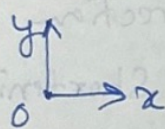
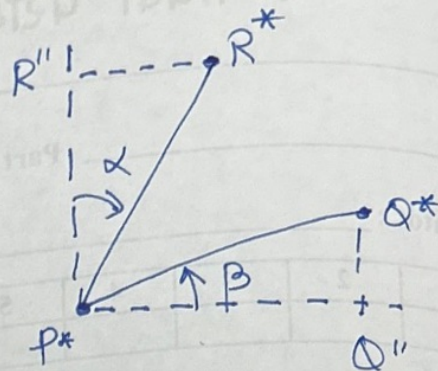
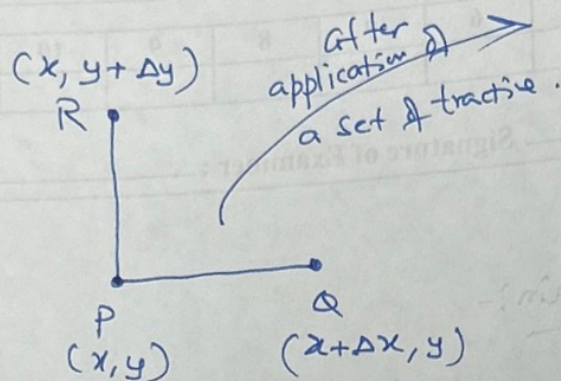
(i) Definition of Normal Strain:-

The normal strain for a given direction (say the x -direction) is the fraction shortening of a line that is originally oriented in the said direction (say the x -direction).

Definition of Shear Strain

Shear strain (ϵ_{xy}) w.r.t the xy -coordinate system is defined as half the increase in angle formed by two infinitesimal line segments that are initially parallel to the x & y -axes.

~~(ii)~~



The co-ordinates of P^* , Q^* & R^* are given by

$$P^* = P - u(P) = (x, y) - [u(x, y), v(x, y)]$$

$$= [x - u(x, y), y - v(x, y)] \quad \text{--- (1)}$$

$$Q^* = Q - u(Q) = (x + \Delta x, y) - [u(x + \Delta x, y), v(x + \Delta x, y)]$$

$$= \{ [x + \Delta x - u(x + \Delta x, y)], [y - v(x + \Delta x, y)] \}$$

→ (2)

$$R^* = R - u(R) = (x, y + \Delta y) - [u(x, y + \Delta y), v(x, y + \Delta y)]$$

$$= [x - u(x, y + \Delta y), y + \Delta y - v(x, y + \Delta y)]$$

→ (3)

Here, Q'' = projection of Q^* on to x-axis

R'' = projecting of R^* on to y-axis.

(1) This means Q'' will have same x-component as Q^*

Similarly R'' will have same y-component as R^* .

Expressing each of the displacements appearing in P^* , Q^* & R^* as Taylor's Series and ignoring terms higher than the 1st order.

$$P^* = (x - u, y - v) \quad \text{--- (4)}$$

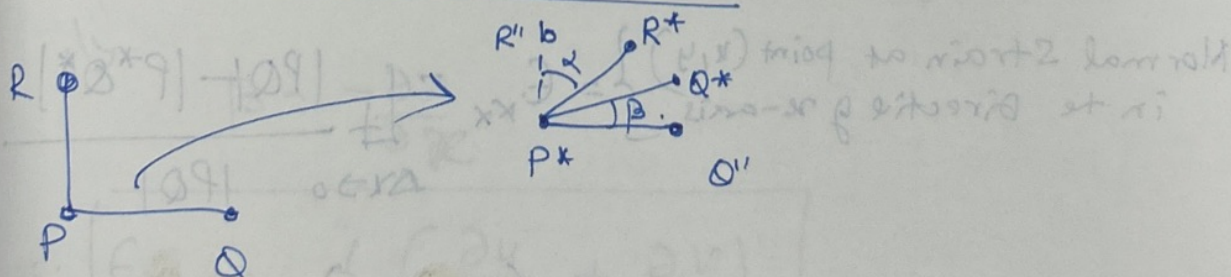
$$Q^* = \left(x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v - \frac{\partial v}{\partial x} \Delta x \right) \quad \text{--- (5)}$$

$$R^* = \left(x - u - \frac{\partial u}{\partial y} \Delta y, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y \right) \quad \text{--- (6)}$$

$$Q'' = \left(x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v \right) \quad \text{--- (7)}$$

$$R'' = \left(x - u, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y \right) \quad \text{--- (8)}$$

Now for. Calculating Normal Strain



Here initial length of $PQ = \Delta x$

& Final Length of $PQ = P^*Q^*$

But $|P^*Q^*|^2 = |P^*Q''|^2 + |Q''Q^*|^2$

$$\text{or, } |P^*Q^*|^2 = \left(\Delta x - \frac{\partial u}{\partial x} \Delta x \right)^2 + \left(\frac{\partial v}{\partial x} \Delta x \right)^2$$

$$= (\Delta x)^2 \left[1 - 2 \frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right]$$

Neglecting the 2nd order partial derivative.

$$|P^*Q^*|^2 = (\Delta x)^2 \left[1 - 2 \frac{\partial u}{\partial x} \right]$$

$$\text{or, } P^*Q^* = \Delta x \cdot \left(1 - 2 \frac{\partial u}{\partial x} \right)^{1/2}$$

using binomial
Theorem &
neglecting higher
terms =

$$|P^*Q^*| = \Delta x \left(1 - \frac{\partial u}{\partial x} \right)$$

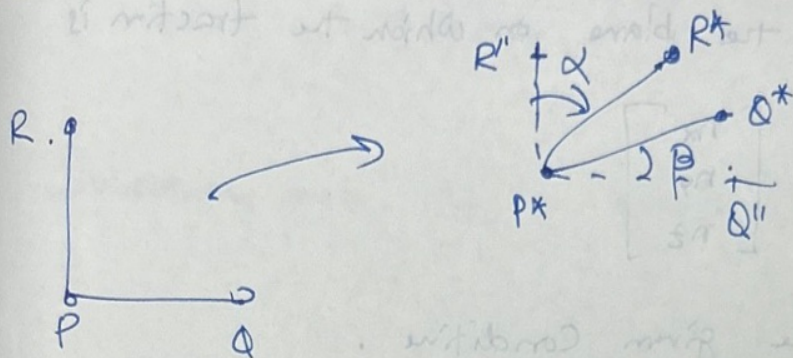
$$\left. \begin{array}{l} \text{Normal Strain at point } (x, y) \\ \text{in the direction of } x\text{-axis} \end{array} \right\} = \epsilon_{xx} = \lim_{\Delta x \rightarrow 0} \frac{|PQ| - |P^*Q^*|}{|PQ|}$$

$$\text{or, } \epsilon_{xx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x - \Delta x \left(1 - \frac{\partial u}{\partial x} \right)}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\therefore \boxed{\epsilon_{xx} = \frac{\partial u}{\partial x}}$$

For. Shear Strain in 2D.

(17)



$$\epsilon_{xy} = \frac{1}{2} L_t \left(\angle R^* P^* Q^* - \angle R P Q \right)$$

$$= \frac{L_t}{\Delta x, \Delta y \rightarrow 0} (90 - \alpha - \beta - 90) = \frac{L_t}{\Delta x, \Delta y \rightarrow 0} (-\alpha - \beta)$$

Now $\tan \alpha = \frac{|R^* R''|}{|P^* R''|} = \frac{-\frac{\partial u}{\partial y} \Delta y}{\Delta y (1 - \frac{\partial v}{\partial y})}$

$$= -\frac{\partial u}{\partial y} \left(1 - \frac{\partial v}{\partial y} \right)^{-1} \rightarrow \text{Binomial Theorem}$$

$$= -\frac{\partial u}{\partial y} \left(1 + \frac{\partial v}{\partial y} \right) = -\frac{\partial u}{\partial y} \quad \text{Neglecting higher order terms}$$

1/y

$$\beta = -\frac{\partial v}{\partial x}$$

$$\therefore \boxed{\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)}$$

Proof

(51)

(18)

(b) Let $\hat{n}$ be the plane on which the traction is zero

$$\hat{n} = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$$

$\therefore$ According to the given condition.

$$\text{The traction } \vec{p}(\hat{n}) = 0$$

$$\Rightarrow \tau \hat{n} = 0$$

$$\Rightarrow \begin{bmatrix} \sigma_x & 20 & 30 \\ 20 & 50 & 15 \\ 30 & 15 & -40 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = 0 \quad \text{--- (1)}$$

Here n_x, n_y & n_z are non-zero. Therefore

$$\begin{vmatrix} \sigma_x & 20 & 30 \\ 20 & 50 & 15 \\ 30 & 15 & -40 \end{vmatrix} = 0$$

$$\sigma_x \begin{vmatrix} 50 & 15 \\ 15 & -40 \end{vmatrix} - 20 \begin{vmatrix} 20 & 15 \\ 30 & -40 \end{vmatrix} + 30 \begin{vmatrix} 20 & 50 \\ 30 & 15 \end{vmatrix} = 0$$

$$\text{or, } \sigma_x (-2000 - 225) - 20 (-800 - 450) + 30 (300 - 1500) = 0$$

$$- 2225 \sigma_x + 20 \times 1250 + 30 \times (-1200) = 0$$

(02)

(19)

$$\sigma_x = - \left(\frac{11,000}{2225} \right) = -4.94 \text{ MPa.}$$

Substituting back σ_x in Equation ① we get

$$-4.94 n_x + 20 n_y + 30 n_z = 0$$

$$20 n_x + 50 n_y + 15 n_z = 0$$

$$30 n_x + 15 n_y - 40 n_z = 0$$

$$\left[\begin{array}{ccc|c} 1 & -4.05 & -6.07 & 0 \\ 1 & 2.50 & 0.75 & 0 \\ 1 & 0.5 & -1.33 & 0 \end{array} \right]$$

$$R_1 \rightarrow \frac{R_1}{-4.94}$$

$$R_2 \rightarrow R_2/20$$

$$R_3 \rightarrow R_3/30$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & -4.05 & -6.07 & 0 \\ 1 & 2.50 & 0.75 & 0 \\ 0 & -2 & -2.08 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & -4.05 & -6.07 & 0 \\ 0 & 6.55 & 6.82 & 0 \\ 0 & -2 & -2.08 & 0 \end{array} \right]$$

$$R_2 \rightarrow \frac{R_2}{6.55}$$

$$R_3 \rightarrow \frac{R_3}{-2}$$

$$\left[\begin{array}{ccc|c} 1 & -4.05 & -6.07 & 0 \\ 0 & 1 & 1.04 & 0 \\ 0 & 1 & -1.04 & 0 \end{array} \right]$$

(P1)

(20)

one of the 3 equations is redundant

unit
 ∴ The vector after normalization

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.789 \\ -0.443 \\ +0.425 \end{bmatrix}$$

Ans

$$\begin{bmatrix} 0 & -1.02 & 1 \\ 0 & 0.92 & 1 \\ 0 & -1.33 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1.02 & 1 \\ 0 & 0.92 & 1 \\ 0 & -1.33 & 1 \end{bmatrix}$$